MIL-P-17578A(SHIPS)
12 APRIL 1967
SUPERSEDING
MIL-P-17578(SHIPS)
9 MAY 1953
(See 6.5)

MILITARY SPECIFICATION

PACKING MATERIAL, PLASTIC (FOR DOGS FOR WATERTIGHT CLOSURES)

(SYMBOL 1425)

1. SCOPE

1.1 Scope.- This specification covers plastic packing material (symbol 1425) consisting of a highly viscous oily binder with a finely ground lubricant filler.

1.2 Classification.- Plastic packing material shall be of the following types, as specified (see 6.2):

Type I - Sticks.
Type II - Coiled strings.
Type III - Bulk.
Type IV - Preformed.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-S-766 - Steel Plate, Sheet, and Strip - Corrosion Resisting.

MILITARY
MIL-P-116 - Preservation, Methods of.

DRAWING

MILITARY
805-1400067 - Doors W. T. Individually Dogged Medium Steel and CRES Dog Details.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 Qualification.- Packing material furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

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3.2 Material.- Plastic packing material shall consist of a highly viscous oily binder and a lubricant filler of finely ground substances. The packing shall be free from all ingredients which have a deleterious effect on corrosion-resisting steel. The ingredients shall be suitable for use at temperatures from 15°F. to 200°F. and pressure differential up to 45 pounds per square inch.

3.3 General.- The packing material shall be permanently homogeneous and plastic and have a consistency proper for handling, lubricating, and sealing purposes.

3.4 Composition and construction.- The composition and construction shall conform to that of the sample submitted for qualification (see 4.2).

3.4.1 Type I (sticks).- Unless otherwise specified (see 6.2), type I packing shall be furnished in the form of sticks approximately 1-1/4 inches long and a diameter of 0.24 plus or minus 0.04 inch.

3.4.2 Type II (coiled strings).- Type II packing shall be similar in composition to type I, but enclosed in a lattice of cotton thread to form a continuous string 0.125 plus or minus 0.03 inch in diameter.

3.4.3 Type III (bulk).- Type III packing shall conform to type I packing but packaged in bulk.

3.4.4 Type IV (preformed).- Type IV packing shall conform to type II packing, but preformed to the dimensions and shape specified (see 6.2).

3.5 Flexibility (types I and III only).-

3.5.1 Type I.- The sticks shall not break when tested in accordance with 4.5.1.

3.5.2 Type III.- The bulk material when formed into sticks of the dimensions given in 3.4.1 and tested in accordance with 4.5.1 shall not break.

3.6 Simulated performance.- During each of the three consecutive cycling operations described in 4.5.2 the packing shall not leak more than 0.20 cubic feet of air and shall not increase the torque required to start rotating the handle (not contacting wedge) more than 10 foot-pounds.

3.7 Corrosion resistance.- There shall be no corrosion of corrosion-resisting steel plates under the packing material when tested as specified in 4.5.3.

3.8 Workmanship.- The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.5. Qualification approval of type I and type II material may also serve as qualification approval for type III and type IV material, respectively.

4.3 Sampling.-

4.3.1 Lot.- For purposes of sampling, a lot shall consist of all packing material of one type, produced in one plant under essentially the same conditions and offered for delivery at one time.

4.3.2 Sampling for examination.- A random sample of units of packing material shall be selected from each lot in accordance with the sampling plan specified in table I for the examination specified in 4.4.1.

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Table I - Sampling for examination.

Number of units 1/ of packing in lot	Number of units of packing in sample	Number of defective units	
		Accept	Reject
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1300	75	4	5
1301 and over	110	6	7

1/ Units of packing - Type I, package, type II, spool, type III, can, type IV, set (see 6.2).

4.3.3 Sampling for comparison tests - From the first lot of packing material of each type offered for delivery under a contract or order, a representative one-pound sample shall be selected. Thenceforth, a representative one-pound sample shall be selected from one of every ten lots which have passed examination The sample so selected shall be forwarded to a laboratory satisfactory to the Naval Ship Engineering Center for tests specified in 4.4.2.

4 4 Examination and tests -

4.4.1 Examination.- Each of the sample units selected in accordance with 4 3 2 shall be examined for workmanship, visual defects, construction, dimensions and tolerances and all other requirements of this specification which do not required tests. Minor surface defects not affecting the serviceability of the packing material shall not be cause for rejection. If any unit of packing material in the sample contains one or more defects, this shall be cause for rejection, and if the number of defective units of packing material in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Comparison tests - The sample selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.5.1 and 4.5.2.

4.4.2.1 Action in case of failure.- Acceptance of the first lot offered for delivery under a contract or order shall be withheld until a satisfactory report is received on the comparison test sample. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the examination specified in 4.4.1. Lot acceptance shall then not be withheld pending receipt of a test report on the subsequent production samples. However, upon receipt of an unsatisfactory test report on a comparison test sample, additional samples from every subsequent lot offered for delivery shall be selected. The sample so selected shall be submitted to a laboratory satisfactory to Naval Ship Engineering Center and shall there be subjected to the comparison tests specified in 4.4.2. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. Additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted.

4.5 Test procedures.-

4.5 1 Flexibility.- The specimen shall be bent half way around a half-inch diameter mandrel in two seconds. The specimen, mandrel, handling equipment, and ambient temperature shall be 23° ± 1 1°C (73.5° ± 2°F.).

4.5.2 Simulated performance.- The wrench end of a Navy type door dog conforming to 805-1400067 shall be enclosed in an air tank and the dog shall be thoroughly packed with the packing material as shown in 805-1400067 Type II (coiled string) and type IV (performed) material may be lubricated and sealed with type I (sticks) or type III (bulk), if necessary. The size of the test tank is optional, but a tank approximately one cubic foot in volume is practical and effective.

4.5.2.1 Test procedure - The dog shall be operated at a rate of 16 plus or minus 2 cycles per minute against air at a gauge pressure of 40 to 45 pounds per square inch (psi) when in equilibrium with an ambient

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temperature of 23° ± 1.1°C. (73.5° ± 2°F.) for the specified number of cycles after each specified preconditioning in the following sequence

Preconditioning	Number of cycles
None (as installed)	400
2 hours at 93.3° ± 1.1°C. (200° ± 2°F.)1/	70
2 hours at minus 9.4° ± 1.1°C. (15° ± 2°F.)1/	30

1/ The packing plunger may be adjusted when necessary, but no additional packing material may be added.

4.5.2.2 Torque measurements.- Torque necessary to start rotation of the handle shall be measured without and then with packing in place, and the difference in foot-pounds reported. The position of the handle shall be the same for all measurements The torque shall be determined as the handle starts to move in both the first and last cycle of each conditioning series. A hand spring scale is suitable for measuring the torque.

4.5.2.3 Air leakage measurement.- Leakage may be calculated from the tank size and pressure drop as follows

$$Q = 0.0634\ VP.$$

Where Q = leakage in cubic feet of air at 23° ± 1.1°C. (73.5° ± 2°F.) and corrected to one atmosphere.
V = Volume of tank in cubic feet.
P = Pressure loss in pounds per square inch.

4.5.3 Corrosion resistence.- The packing material shall be applied in streaks about half an inch apart on 3-inch square plates of corrosion-resisting steel conforming to class 304 of QQ-S-766. The plates shall be exposed to a 20 percent sodium chloride spray at 35° ± 1.1°C. (95° ± 2°F.) for 400 ± 1/4 hours continuously. After this exposure period the steel under the packing strips shall be examined for corrosion.

4.6 Examination of preparation for delivery.- An examination shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Examination of preparation for delivery shall be in accordance with the following. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-P-116. The Acceptable Quality Level (AQL) shall be 2.5 percent.

Examine	Defect
Packaging	Unit package not packaged as specified. Level not as specified. Packaging material not as specified. Closure not as specified.
Packing	Not as specified. Container not as specified; closure not accomplished by specified or required methods or materials. Any nonconforming component, component missing, damaged, or otherwise defective, affecting serviceability. Inadequate application of components such as incomplete closure of case-liners, container flaps, loose or inadequate strappings; bulged or distorted containers.
Count	Number of items per container less than specified or indicated quantity.
Weight: Fiberboard cartons	Gross weight exceeds specified requirements.

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Examine	Defect
Weight: (Cont'd)	
Wood or wood-cleated boxes	Gross weight exceeds 200 pounds.
Markings	Interior or exterior markings (as applicable), omitted, illegible, incorrect, incomplete, or not as specified.

4.6.1 If the number of defective units exceeds the AQL, this shall be cause for rejection of the entire lot represented by the units.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.-

5.1.1 Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal Stock number or manufacturer's part number, contract or order number, contractor's name and destination, and date of manufacture.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Packaging.- Packaging shall be Level A or C, as specified (see 6.2).

5.2.1.1.1 Level A.-

5.2.1.1.1.1 Type I.- Type I (sticks) packing material shall be wrapped or suitably enclosed in greaseproof barrier material conforming to grade A of MIL-B-121. The packing material shall then be placed in snug fitting unit and intermediate containers conforming to PPP-B-566, PPP-B-636, or PPP-B-676. Container closure and sealing shall be in accordance with the applicable container specification or Appendix thereto.

5.2.1.1.1.2 Type II.- Type II (coiled strings) shall be furnished on commercial metal spools. The spools shall be wrapped or suitably enclosed in greaseproof barrier material as specified in 5.2.1.1.1.1 and packaged in containers as specified in 5.2.1.1.1.1.

5.2.1.1.1.3 Type III.- Type III (bulk) packing material shall be furnished in 5-pound metal cans conforming to type V, class 2 or 4 of PPP-C-96. Exterior plan B coating and side seam striping shall be required. Cans shall be provided with wire handles which shall be galvanized or otherwise protectively coated to resist corrosion.

5.2.1.1.1.4 Type IV.- Type IV (preformed) shall be wrapped or suitably enclosed in greaseproof barrier material conforming to grade A of MIL-B-121. The packing material shall then be placed in snug fitting tube containers conforming to PPP-C-55. When specified (see 6.2), only one set of packing shall be packaged

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in one container. Tube containers shall then be placed in unit or intermediate containers conforming to PPP-B-566, PPP-B-636, or PPP-B-676.

5.2.1.1.2 Level C.- Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for early use and may conform to the supplier's commercial practice.

5.2.1.2 Packing.- Packing shall be Level A or B, as specified (see 6.2).

5.2.1.2.1 Levels A and B.-

5.2.1.2.1.1 Types I, II and IV.- Types I, II and IV packing material, packaged as specified (see 6.2), shall be packed in containers conforming to class weather resistant of PPP-B-636 or class 2 of PPP-B-640, at the option of the supplier. All corners and edge seams and manufacturer's joints of fiberboard boxes shall be waterproofed with tape in accordance with the Appendix to the box specification. Boxes shall be closed and banded with pressure sensitive tape in accordance with the applicable box specification or Appendix thereto. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Fiberboard containers when used for packaging (see 5.2.1.1.1.1 and 5.2.1.1.1.4), conforming to class weather resistant of PPP-B-636, closed, sealed and banded as specified herein may be used as the shipping container and need not be overpacked.

5.2.1.2.1.2 Type III.- Type III packing material packaged in cans shall be arranged and packed for overseas shipments in accordance with the requirements of the Appendix to PPP-C-96.

5.2.1.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture.)

6. NOTES

6.1 Intended use.- Plastic packing material (symbol 1425) is intended for use in packing dogs and operating handwheel spindles of watertight doors.

6.2 Ordering data.- Procurement documents should specify the following

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Dimensions and shape (see 3.4.1 and 3.4.4).
(d) Number of type IV (preformed) packing constituting one set (see 4.3.2).
(e) Packaging, packing or marking requirements other than those required by 5.1 (see 5.2).
(f) Packaging only one set of type IV packing in one container, when required (see 5.2.1.1.1.4).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-17578 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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6.4 The performance of type II (coiled strings) is improved by installing the packing in a solid coil on the low pressure side of the spring instead of in the bifilar configuration (alternating with the spring coils) as shown in 805-1400067.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-17578A(SHIPS)	PACKING MATERIAL, PLASTIC (FOR DOGS FOR WATERTIGHT CLOSURES) (SYMBOL 1425)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE
OF REINSTATEMENT

MIL-P-17578A(SHIPS)
NOTICE 2
7 June 1985
SUPERSEDING
NOTICE 1
10 December 1984

MILITARY SPECIFICATION

PACKING MATERIAL, PLASTIC (FOR DOGS FOR WATERTIGHT CLOSURES) (SYMBOL 1425)

Specification MIL-P-17578A(SHIPS), dated 12 April 1967, is hereby reinstated and may be used for acquisition.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Preparing activity:
Navy - SH
(Project 5340-N095)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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NOTICE OF VALIDATION

MIL-P-17578A(SHIPS)
NOTICE 3
27 March 1991

MILITARY SPECIFICATION

PACKING MATERIAL, PLASTIC (FOR DOGS FOR WATERTIGHT CLOSURES) (SYMBOL 1425)

MIL-P-17578A(SHIPS), dated 12 April 1967, has been reviewed and determined to be valid for use in acquisition.

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